Niranjan K R

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Professional Summary :

A motivated BTech Biotechnology student with a strong interest in programming, seeking to learn, grow, and contribute through both biological and computational skills. To build a career that combines biotechnology and coding, by applying programming skills to solve real-world scientific and technical problems. To secure an opportunity where I can apply my knowledge of biotechnology along with my growing interest in coding, and continuously develop my technical and analytical skills.

Internships

SRI SHAKTHI FOOD TESTING LABORATORY

Food, water and soil analysis both in chemical and microbiology lab | 02.09.2025 - 16.09.2025

- Learned food safety rules, lab SOPs, and hygiene practices at Sree Sakthi Food Testing Lab.
- Gained practical experience in microbiological testing of food and water samples.
- Learned chemical, water, and soil analysis using lab instruments and titration methods.

BIOXPLORA

Molecular docking and drug discovery studies | 13.10.2025 - 31.10.2025

- Understood the basics of molecular docking and its role in drug discovery.
- Learned to use biological databases to retrieve protein and ligand information.
- Gained exposure to various software applications for docking, analysis, and interpretation of results.

VJ BIOTECH

Bio-Pharma, nanotechnology and chromatographic techniques | 05.01.2026 - 24.01.2026

- Learned basic chromatography and separation techniques.
- Performed nanoparticle synthesis and biological activity tests.
- Gained knowledge in bio-pharma and natural product studies.

THANGAM INFO TECH

Full-Stack Development | 04.02.2026 - 04.04.2026

- Contributed to building a Clinical Management System for handling patient records using React.js, TypeScript, Tailwind CSS, .NET Core Web API, and MS SQL Server.
- Developed and implemented RESTful APIs and CRUD operations to ensure smooth data processing and system functionality.
- Integrated database systems to improve data accuracy and optimize overall performance and scalability.

SREE MAHALAKSHMI TECHNOLOGY SOLUTIONS

Artificial Intelligence and Machine Learning | 02.03.2026 - 25.03.2026

- Collected and analyzed datasets for model development.
- Built and tested machine learning models using Python.
- Practiced coding and algorithms on Kaggle and HackerRank.

Projects

Microbial Diversity Analysis of Microbial Taxa Associated with Pork Intramuscular Lumen

- This study focuses on identifying bacteria present in the intramuscular lumen of pork using culturing techniques and 16S rRNA gene sequencing.
- The identified microbial sequences are analyzed and submitted to the NCBI database to understand microbial diversity and uniqueness.

Phytochemical Analysis and Antidiabetic Potential of Syzygium cumini

- This study identifies major phytochemicals present in Syzygium cumini seed extract using standard screening methods.
- The antidiabetic potential is evaluated through alpha-glucosidase enzyme inhibition along with antimicrobial and antioxidant assays.

Dye Degradation Using Honey-Mediated Silver Nanoparticles

- Silver nanoparticles are synthesized using natural honey as a green, eco-friendly reducing and stabilizing agent.
- The photocatalytic efficiency of the synthesized AgNPs is evaluated for sunlight-driven degradation of methylene blue dye in wastewater.

Drug Consumption Prediction using Machine Learning

- Built a machine learning model to predict drug usage behavior using demographic and psychological data.
- Implemented and compared algorithms like Logistic Regression, Decision Tree, and Random Forest to identify the best-performing model.

Full-Stack Application - Clinical Management System

- Developed a Clinical Management System to manage patient records efficiently using a modern full-stack architecture.
- Built a responsive React frontend with type-safe patterns and API-based CRUD operations for seamless data management.

Key Skills

Biotechnology & Lab

Bioinformatics • Wet lab • Experimental biotechnology • Analytical techniques

Frontend

React • TypeScript • JavaScript • HTML • CSS • Tailwind CSS • Responsive UI • Lifecycle Hooks

Backend & APIs

.NET Core Web API • Node.js • JWT Authentication • API Integration

Database

SQL Server • MySQL • MongoDB • UCI repo

Programming Languages:

C • Python

AI skills

Algorithms implementation, Problem-solving, AI tools

Tools & Methodologies

Git • Git Version Control (branching, merging, pull requests) • GitHub • VS Code • Visual Studio • PowerBI • UiPath Studio • Excel

Education

Bachelor of Technology (B.Tech.) – Biotechnology (2024 – 2028) **CGPA – 8.415**

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